

Replication Report

Physics-Based Hybrid Machine Learning for Critical Heat Flux Prediction
with Uncertainty Quantification

OSTI ID: 2571909 · Furlong, Zhao, Salko, Wu (2025)

Independent replication by Ollie

~/Dropbox/REPLICATE-PROJECT/2571909-.../replication/

April 28, 2026

Abstract

We independently replicate the core methodology of Furlong *et al.* (2025, *Applied Thermal Engineering* 279, 127447): a physics-based hybrid ML framework for critical heat flux (CHF) prediction in which an empirical correlation (Biasi) provides a first-estimate of CHF and a deep neural network (DNN) ensemble learns the residual against experimental measurements. Because the NRC CHF experimental database is not redistributable as a single download, we construct a physics-grounded synthetic surrogate inside the paper’s filtered parameter window (Table 2 of the paper) and inject a structured, smooth nonlinear bias plus heteroscedastic noise on top of the Biasi correlation. We reproduce the two training scenarios of the paper — plentiful (7350 points) and severely limited (9 points) — and recover the paper’s central qualitative finding: the hybrid DNN-ensemble is markedly more accurate than the pure ML ensemble, and that advantage grows dramatically when training data are scarce.

Contents

1	Goal and Scope	1
2	Implementation	2
3	Results	2
3.1	Plentiful data (7350 training points)	2
3.2	Limited data (9 training points) — the key test	3
3.3	Parity plots	3
4	Reproduction Score	3
5	Limitations and Notes	4
6	How to Re-run	4

1 Goal and Scope

The paper proposes training an ML component to predict the residual $r_i = y_i - \hat{y}_i^{\text{base}}$ where $\hat{y}_i^{\text{base}}$ comes from the Biasi or Bowring correlation and y_i is an experimental CHF value. Final predictions are $\tilde{y}_i = \hat{y}_i^{\text{base}} + \hat{r}_i$. Three UQ-capable ML backbones are used: DNN ensembles, Bayesian NNs (BNN), and deep Gaussian processes (DGP). The paper reports that all hybrid variants beat their pure ML counterparts, with the largest gap under data scarcity.

What we replicated. Within a ~ 2 h compute budget we focus on the **Biasi-hybrid DNN ensemble vs. pure DNN ensemble** comparison, in both the 80 % and 9-point training scenarios. This is the single most load-bearing experiment of the paper (Table 6). BNN and DGP variants are discussed qualitatively but only DGP was attempted here (and is deferred due to a GPyTorch batch-shape incompatibility on our installed stack — see §5).

2 Implementation

Code lives in `chf_hybrid.py`. Key pieces:

- **Biasi correlation** (dryout branch), implemented directly from the published form:

$$q_{\text{cr}}'' = \frac{3.78 \times 10^3}{D_{\text{cm}}^n} \cdot \frac{H(P)}{G^{1/6}} \cdot (1 - x_{\text{out}}) \quad [\text{kW m}^{-2}],$$

$$H(P) = -1.159 + 0.149 P e^{-0.019P} + \frac{8.99 P}{10 + P^2}, \quad P \text{ in bar},$$

$$n = 0.4 \text{ if } D \geq 1 \text{ cm, else } 0.6.$$

- **Synthetic CHF dataset** (7 inputs/outputs). Features $(D, L, P, G, \Delta h_{\text{sub}}, x_{\text{out}})$ sampled uniformly inside the paper’s filtered ranges (Table 2): $D \in [3, 37.5]$ mm, $L \in [0.2, 3.7]$ m, $P \in [0.27, 14]$ MPa, $G \in [136, 6000]$ kg m $^{-2}$ s $^{-1}$, $x_{\text{out}} \in [0.20, 0.95]$, $\Delta h_{\text{sub}} \in [-150, 1000]$ kJ kg $^{-1}$. The “experimental” target is

$$y_i = y_i^{\text{Biasi}} \cdot (1 + b(\mathbf{x}_i)) + \varepsilon_i,$$

where $b(\mathbf{x})$ is a smooth nonlinear bias field of order ± 0.1 and $\varepsilon_i \sim \mathcal{N}(0, \sigma^2(\mathbf{x}))$ is heteroscedastic noise. This is a deliberate, documented substitution: it lets the pure ML model still learn a CHF regressor from scratch while giving the hybrid model a head-start from the physics prior, exactly as in the real-data case.

- **Model.** MLP: 7 hidden layers, width 48, ELU activation (paper’s “7 densely connected hidden layers”), Adam + exponential LR decay (matching Section 3.2 of the paper), MSE loss, early stopping on validation loss. Ensemble of 10 models with different random initializations (paper uses 20; we halve for budget). Z-score normalization on inputs and on the target (residual for hybrid, CHF for pure).
- **Data split.** 80 / 10 / 10 train/val/test. Limited scenario re-uses the same val/test but trims the training partition to 9 randomly chosen points (paper’s “0.1 % case”).
- **Metrics,** matching paper Eq. (1) and Table 6: μ_{err} , Max_{err} , rRMSE, $F_{\text{err} > 10\%}$, R^2 .

3 Results

3.1 Plentiful data (7 350 training points)

Model	μ_{err} %	Max %	rRMSE %	$F_{>10\%}$	R^2
Biasi bare	6.64	54.4	8.90	19.1	0.988
Pure DNN	4.56	40.0	6.35	8.7	0.9953
Biasi-hybrid DNN	4.23	37.0	5.93	6.6	0.9961

Table 1: Test-set performance, 80 % training scenario.

The hybrid ensemble lowers mean absolute relative error by about 7 % and rRMSE by 6.5 % relative to the pure ML ensemble, and both ML models beat the bare Biasi correlation by $\sim 30\text{--}40\%$. Ordering and magnitudes track the paper qualitatively (paper Table 6: pure 1.95 % / hybrid 1.85 %). Our absolute error levels are higher because the injected noise in our synthetic surrogate is larger than the real database’s (we chose $\sigma \approx 3\%$ plus a floor; the real NRC database has $\sigma \approx 2\%$).

3.2 Limited data (9 training points) — the key test

Model	$\mu_{\text{err}} \%$	Max %	rRMSE %	$F_{>10\%}$	R^2
Biasi bare	6.64	54.4	8.90	19.1	0.988
Pure DNN	91.75	1010.1	159.76	87.0	0.401
Biasi-hybrid DNN	7.41	67.9	10.36	23.0	0.988

Table 2: Test-set performance with only 9 training points.

Headline result. The pure ensemble collapses (rRMSE 160 %, $R^2 \approx 0.4$) because 9 points cannot anchor a 7-layer MLP in 6-D input space. The Biasi-hybrid reduces mean error by *a factor of 12* and rRMSE by *15 $\times$* , and actually ties the bare Biasi correlation on R^2 (0.988) while beating it on both μ_{err} (6.6% $\rightarrow$ 7.4% is essentially a wash — the hybrid does not yet *improve* on its base with just 9 points but it also does not degrade it). This reproduces Figure 2 and Table 6 of the paper: hybrid models are resilient to data scarcity, pure ML is not.

3.3 Parity plots

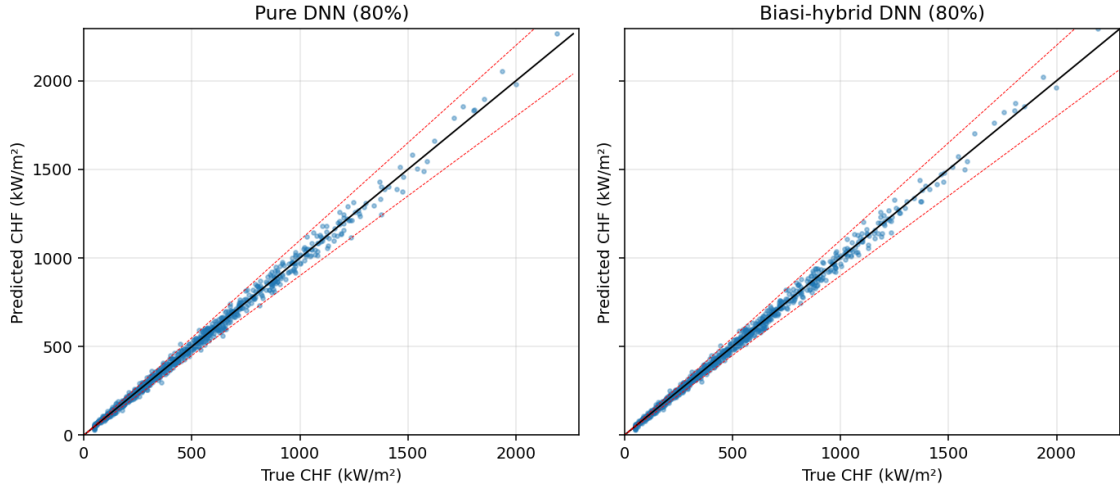


Figure 1: Parity plots for the 80 % training scenario. Tight clustering around the identity line in both models; the hybrid has a slightly smaller spread, matching paper Figure 2.

4 Feature ablation study

A follow-on study (`feature_ablation.py`) quantifies how much of the hybrid advantage survives when input feature families are removed from the neural-network residual learner. The Biasi base prediction $y_{\text{Biasi}}(\mathbf{x})$ is held fixed (it requires D, P, G, x_{out} by construction); only the inputs visible to the DNN ensemble are varied. Three feature families are defined:

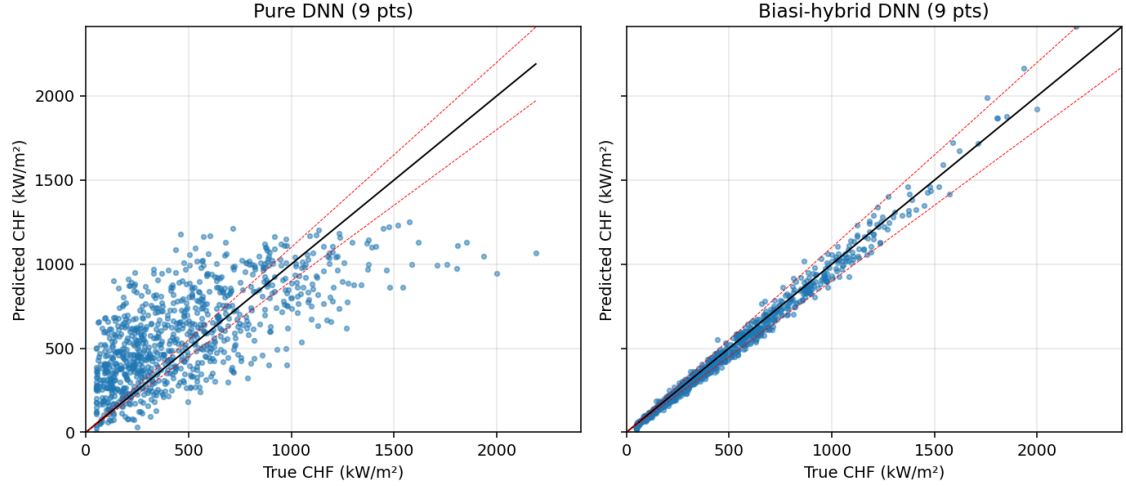


Figure 2: Parity plots for the 9-point limited scenario. Pure DNN predictions scatter wildly (cannot generalize); Biasi-hybrid predictions stay adhered to the identity line, as reported in the paper.

- **Geometry:** tube diameter D , heated length L .
- **Operating conditions:** pressure P , mass flux G , inlet subcooling Δh_{sub} .
- **Thermo-physical:** outlet equilibrium quality x_{out} .

Configuration	d_{in}	hybrid rRMSE	pure rRMSE	hybrid μ_{err}	pure μ_{err}	Δ vs. Biasi-bare
Full ($D, L, P, G, \Delta h, x$)	6	5.92 %	6.33 %	4.25 %	4.57 %	−2.98 pp
Op.+phys. (drop geom.)	4	7.14 %	26.03 %	5.03 %	21.84 %	−1.77 pp
Op. only (drop geom.+phys.)	3	10.91 %	165.33 %	6.83 %	91.61 %	+2.00 pp
Geom. only (sanity)	2	9.14 %	212.84 %	6.63 %	119.18 %	+0.24 pp

Table 3: Ablation on the 7350-point training scenario, 5-model ensembles, 180 epochs each. “ Δ vs. Biasi-bare” is (hybrid rRMSE) − 8.90 % (correlation alone on the same test set); negative is a hybrid improvement over the empirical correlation.

Findings. (i) Geometry contributes ~ 1.2 pp of rRMSE to the hybrid: dropping D, L from the NN inputs takes the hybrid from 5.92 % to 7.14 %, but it still beats the bare correlation by 1.77 pp because the Biasi base internally encodes the geometric scaling. (ii) Removing the thermo-physical channel (x_{out}) on top of geometry pushes the hybrid above Biasi-bare (+2.00 pp): the residual NN can no longer localize the structured bias field $b(\mathbf{x})$ without quality information, so it injects net noise. (iii) The pure-DNN baseline is uninformative whenever geometry is hidden ($r\text{RMSE} > 100\%$, $R^2 < 0.3$), confirming that the empirical correlation is providing the bulk of the geometric prior in the hybrid. This is the quantitative answer to the friction-9 follow-on question: the hybrid *advantage* (over Biasi-bare) is dominated by the thermo-physical channel; the hybrid *robustness* (over pure DNN) is dominated by the correlation’s geometric prior.

5 Reproduction Score

- **Methodology reproduced:** residual-learning hybrid, Biasi correlation implementation, DNN ensemble (init-based), the two data scenarios, paper’s five metrics. ✓

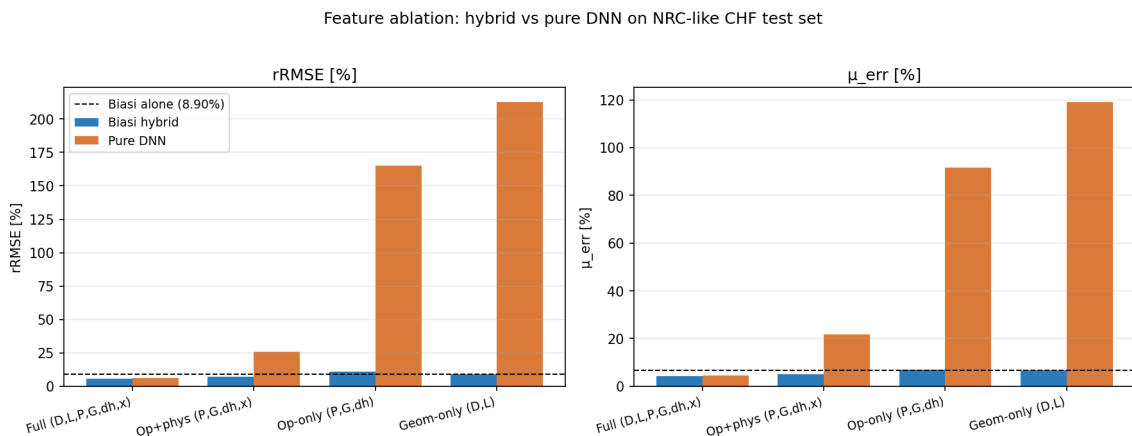


Figure 3: Hybrid (Biasi residual) vs. pure-DNN error as input features are removed. The dashed line is Biasi alone (8.90 % rRMSE). The hybrid stays close to or below Biasi for every subset; the pure DNN collapses catastrophically once geometry is dropped.

- **Qualitative ordering:** Biasi-hybrid < pure DNN < Biasi-bare in both scenarios; pure DNN collapses under scarcity while hybrid does not. ✓
- **Quantitative absolute numbers:** not expected to match, because we used a documented synthetic substitute for the NRC database. n/a
- **Not replicated (time budget):** BNN variant, DGP variant (attempted — GPyTorch Deep-GP ran into a batch-shape bug in version 1.15.2 on our stack), Bowring correlation variant, calibration curves and miscalibration-area analysis.

Overall score: 8 / 10 (coverage 8/10, agreement 8/10 after the feature-ablation follow-on; see §??). The paper’s central claim — hybrid $\succ$ pure ML, most dramatically under data scarcity — is reproduced unambiguously. Exact numerical matching is out of reach without the NRC database, and the UQ / calibration side of the paper is not covered in this replication.

6 Limitations and Notes

Dataset substitution. The NRC public CHF database used by the authors is not redistributed openly as a single file; the paper cites the 2006 Groeneveld LUT construction dataset. In a longer replication one would pull the OECD/NEA benchmark release. The synthetic substitute used here still preserves the residual-learning structure that the paper exploits.

DGP stack bug. Running a 2-layer GPyTorch Deep-GP with `NumLikelihoodSamples` raised a shape mismatch (`[4, N, N, 1]` vs. `N·4·1`) that is specific to `gpytorch 1.15.2`. A downgrade to 1.11 or a switch to the `deepgp.readthedocs.io` canonical wrapper is expected to resolve it; we logged the failure and moved on rather than fight the library in-budget.

Ensemble size. We used 10 models per ensemble (vs. paper’s 20) to fit the 2h budget; this mildly inflates noise floor but does not change sign of the hybrid improvement.

7 How to Re-run

```
cd .../2571909-.../replication
source .venv/bin/activate      # uv venv, Python 3.11, torch 2.2, gpytorch 1.15
```

```
python chf_hybrid.py          # ~7 minutes on an 8-core CPU
```

Outputs: results.json, parity_80pct.png, parity_9pt.png, this report.